

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Javascript

Deep Learning & ML: TensorFlow, PyTorch, Scikit-learn, Predictive Analytics, Machine Learning, Künstliche Intelligenz

MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, Real-time System Optimization

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing, Business Intelligence Concepts

Core Skills: Mathematics, Statistics, Process Analysis, Optimization, Prototyping, Data Visualization

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Developed and evaluated **predictive analytics** and **machine learning** models for metrological applications, enhancing model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to improve **model accuracy and stability**, generating **prototypes** to validate innovative solution approaches.
- Collaborated effectively with PhD researchers and the **BI Team** on complex projects, contributing to a publication and fostering a **team-oriented** environment.
- Designed and implemented dashboards for **data visualization** and comprehensive **model performance monitoring**, enabling data-driven insights.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized **AI algorithms** for vitiligo segmentation, achieving 94% accuracy and 5x speed improvement, demonstrating expertise in **process optimization**. Deployed solutions with Docker and FastAPI.
- Developed a **real-time** tennis foul detection system leveraging TensorRT, reducing compute cost and processing time by 6x, showcasing strong **analytical** and technical problem-solving.
- Managed end-to-end **MLOps pipelines** on AWS SageMaker, proactively addressing data drift and ensuring continuous improvement of **data-driven solutions**. Maintained Kubernetes infrastructure.
- Conducted educational sessions on **Machine Learning** and big data, advising stakeholders and fostering knowledge sharing in a **team-oriented** setting.

- Developed proof-of-concepts (**prototypes**) using YOLO, few-shot learning, and CNN for client demonstrations, translating complex **AI solutions** into tangible benefits.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Executed full-stack development of a website offering **AI-based services**, utilizing Flask, HTML, and CSS, demonstrating comprehensive technical skills.
- Developed an **AI chatbot application** using the OpenAI API and relevant LLM frameworks, focusing on **customer-oriented** solutions and rapid **prototyping**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Developed robust web scrapers and **ETL pipelines** for structured and unstructured data, contributing to efficient **data processing** for **data-driven solutions**.
- Automated **data ingestion** and cleaning workflows, enhancing data quality for client **analytics use cases** and supporting BI system improvements.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble for **protein function prediction**, integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation, showcasing advanced **predictive analytics**.
- Increased **model accuracy** by 4% through custom feature engineering across MF, CC, and BP namespaces, demonstrating strong **analytical** and **optimization** skills.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed an LSTM-based deep learning model for **real-time Bitcoin price forecasting**, achieving high accuracy in **predictive analytics**.
- Applied advanced **mathematical and statistical methods** for time-series analysis to identify **optimization potentials** in financial predictions.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed real-time signal processing system integrating temporal dependencies for music generation
- Implemented transformer-based models for sequence prediction in audio generation tasks

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, demonstrating technical writing and open-source contribution in the **AI** domain.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python